Coms 472/572 Recitation

INFORMED SEARCH
SLIDES PARTIALLY ADAPTED FROM DR. YAN-BIN JIA

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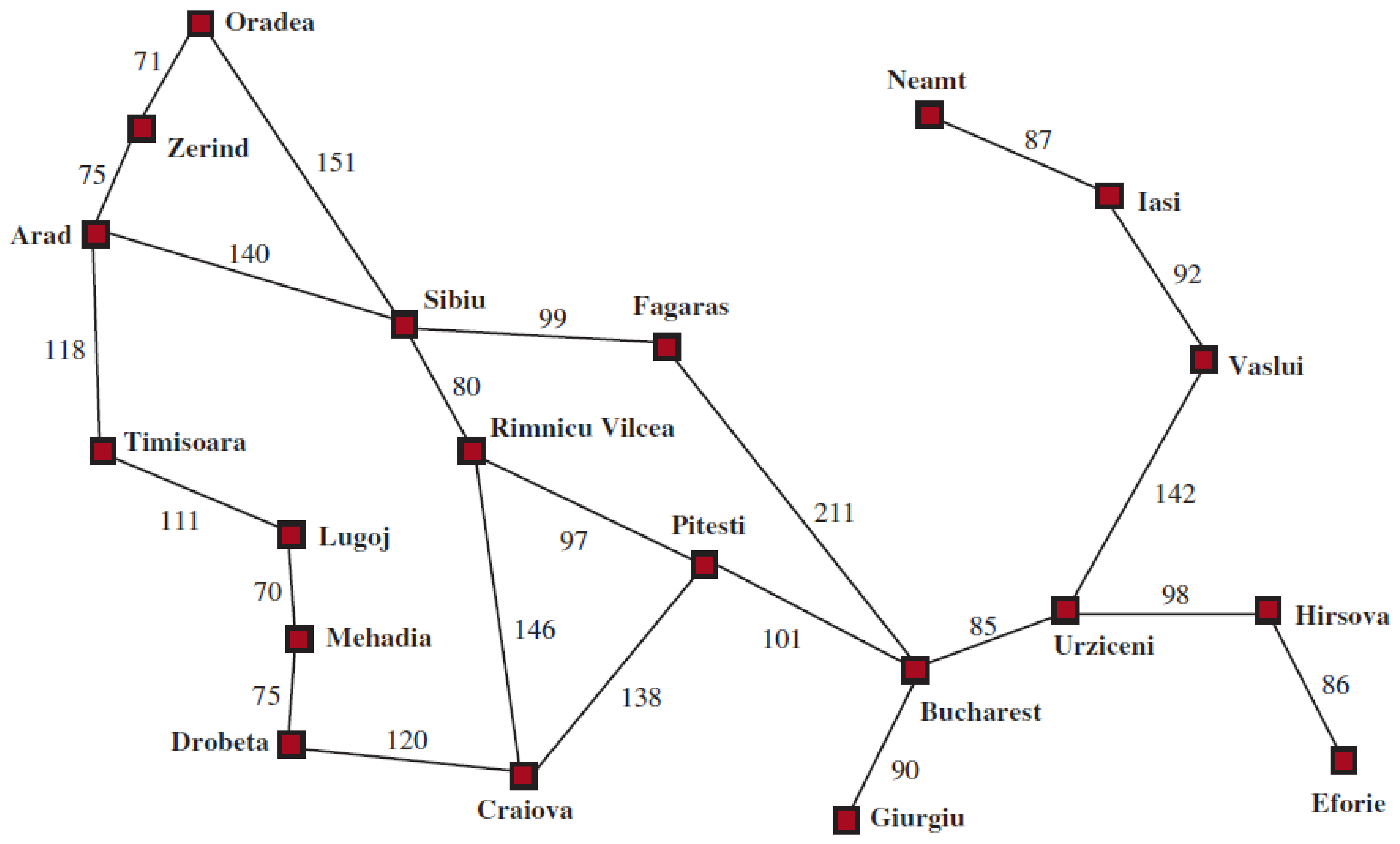
I. Informed (Heuristic) Search

Use of domain-specific hints about the location of a goal can find a solution more efficiently.

Heuristic function:

$h(n)$ = *estimated* cost of the cheapest path from the state at node n to a goal state

e.g., straight-line distance h_{SLD} on the map between two sites



Arad	366	**Mehadia**	241
Bucharest	0	**Neamt**	234
Craiova	160	**Oradea**	380
Drobeta	242	**Pitesti**	100
Eforie	161	**Rimnicu Vilcea**	193
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Straight-line distance to Bucharest

Greedy Best-First Search

Evaluation function $f(n) = h(n)$

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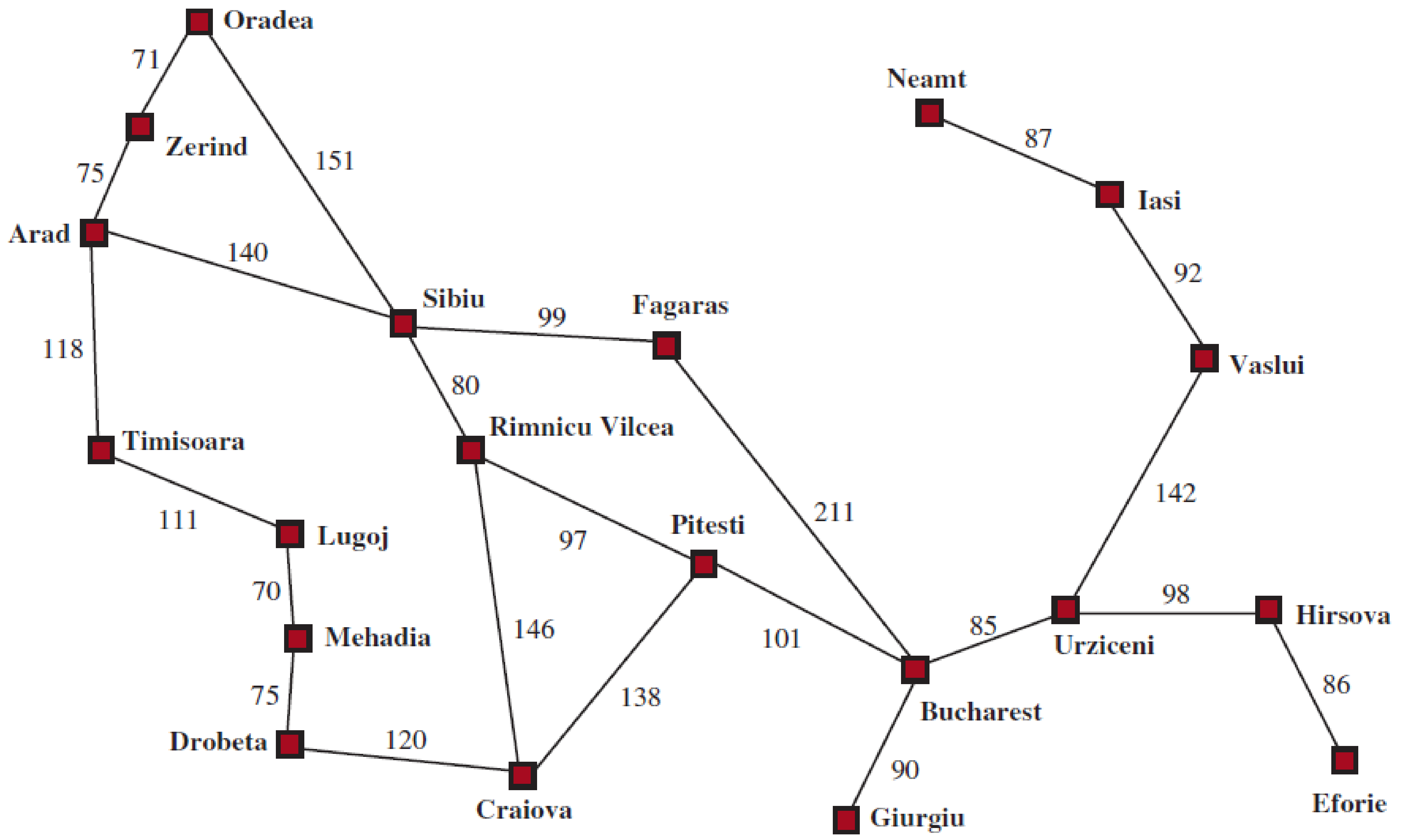
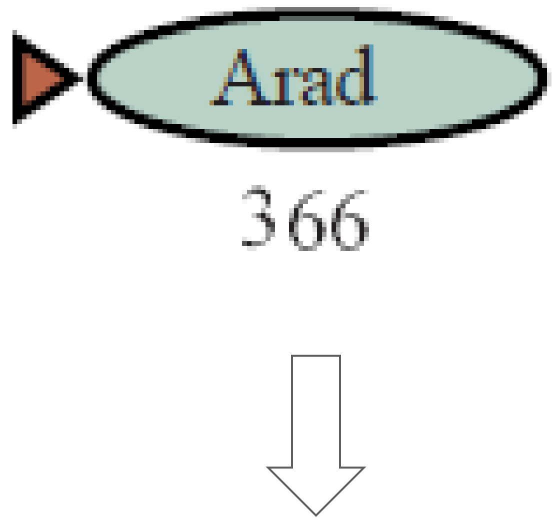
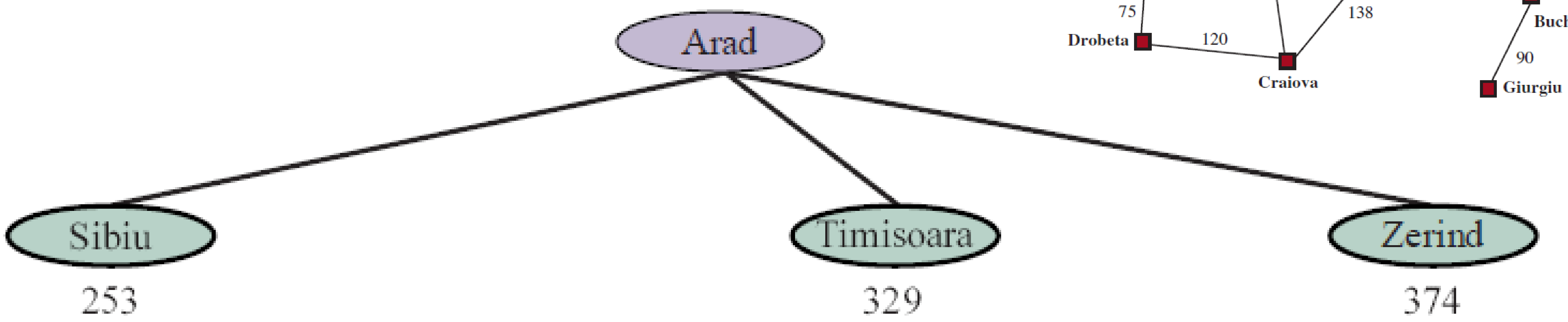
$$f(n) = h_{SLD}(n)$$

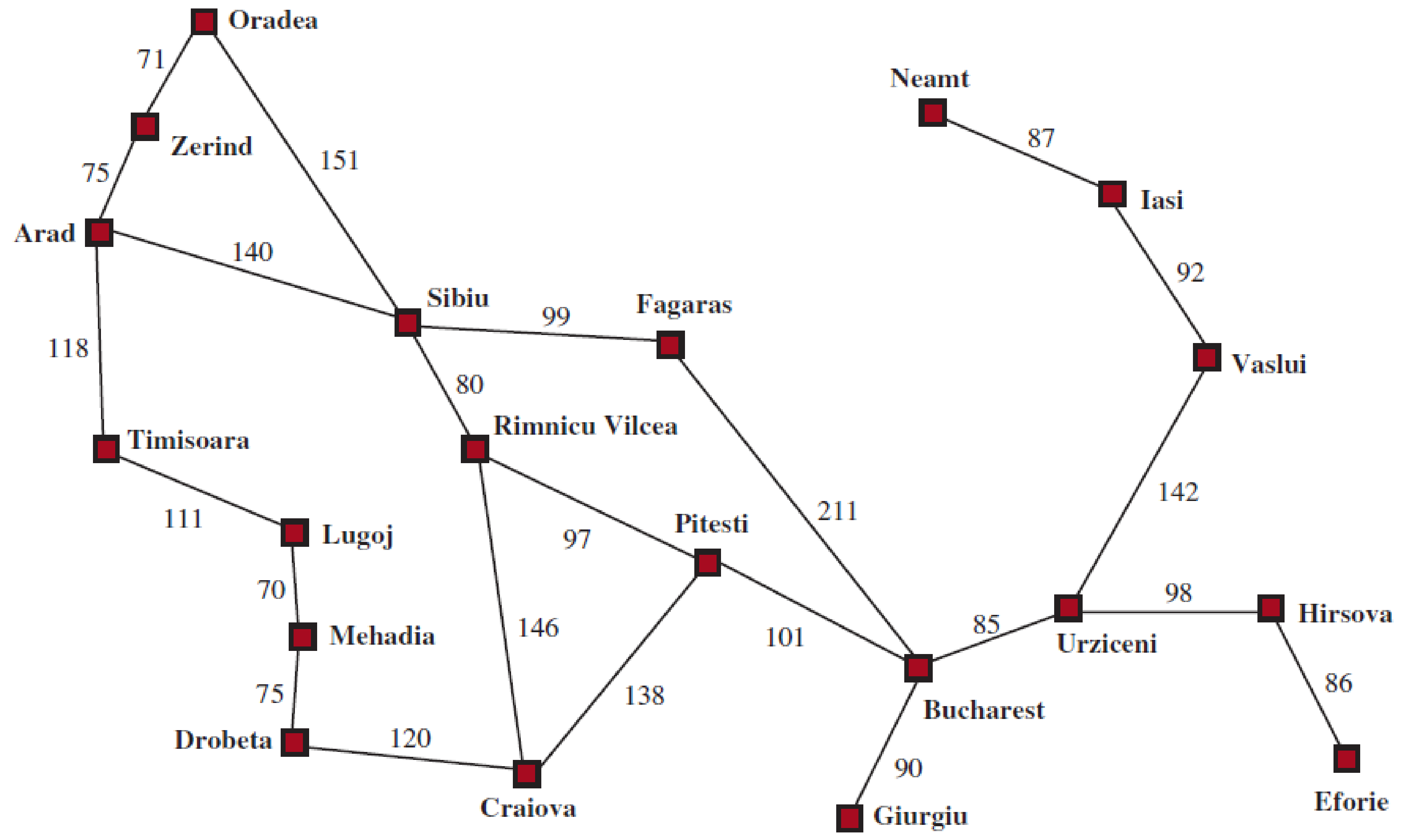
Good correlations with road distances



Greedy Search for Bucharest







Closer to Bucharest
than is Zerind or Timisoara

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Search for Bucharest (cont'd)

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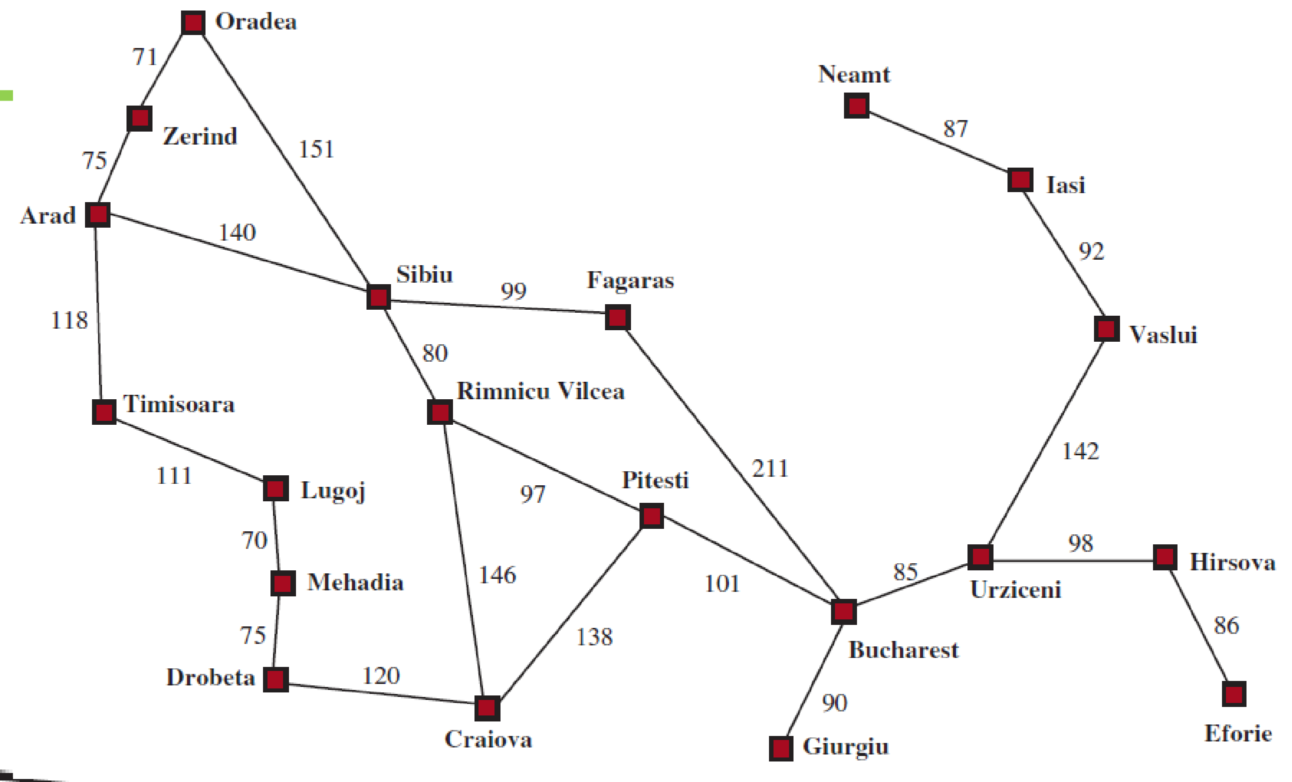
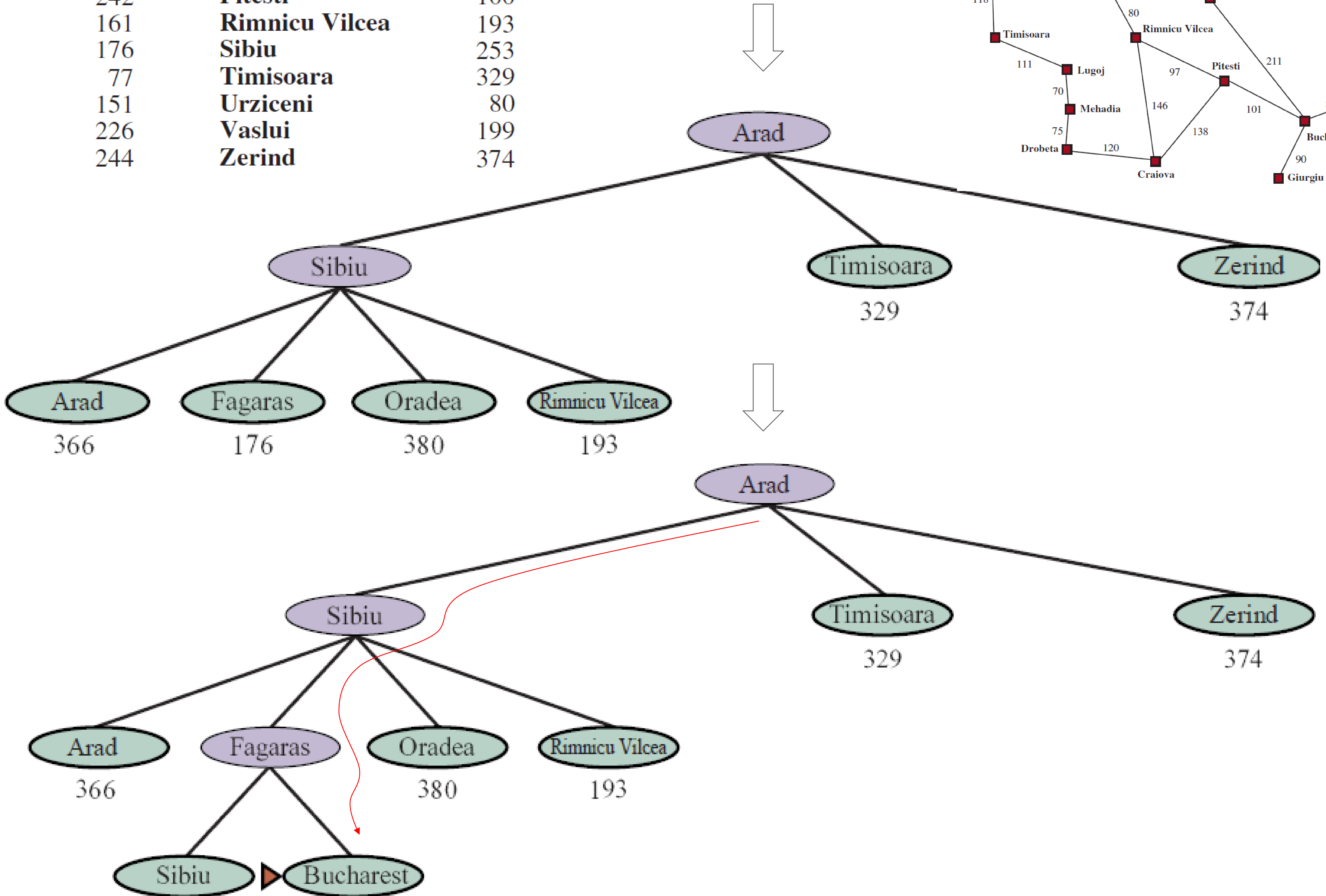
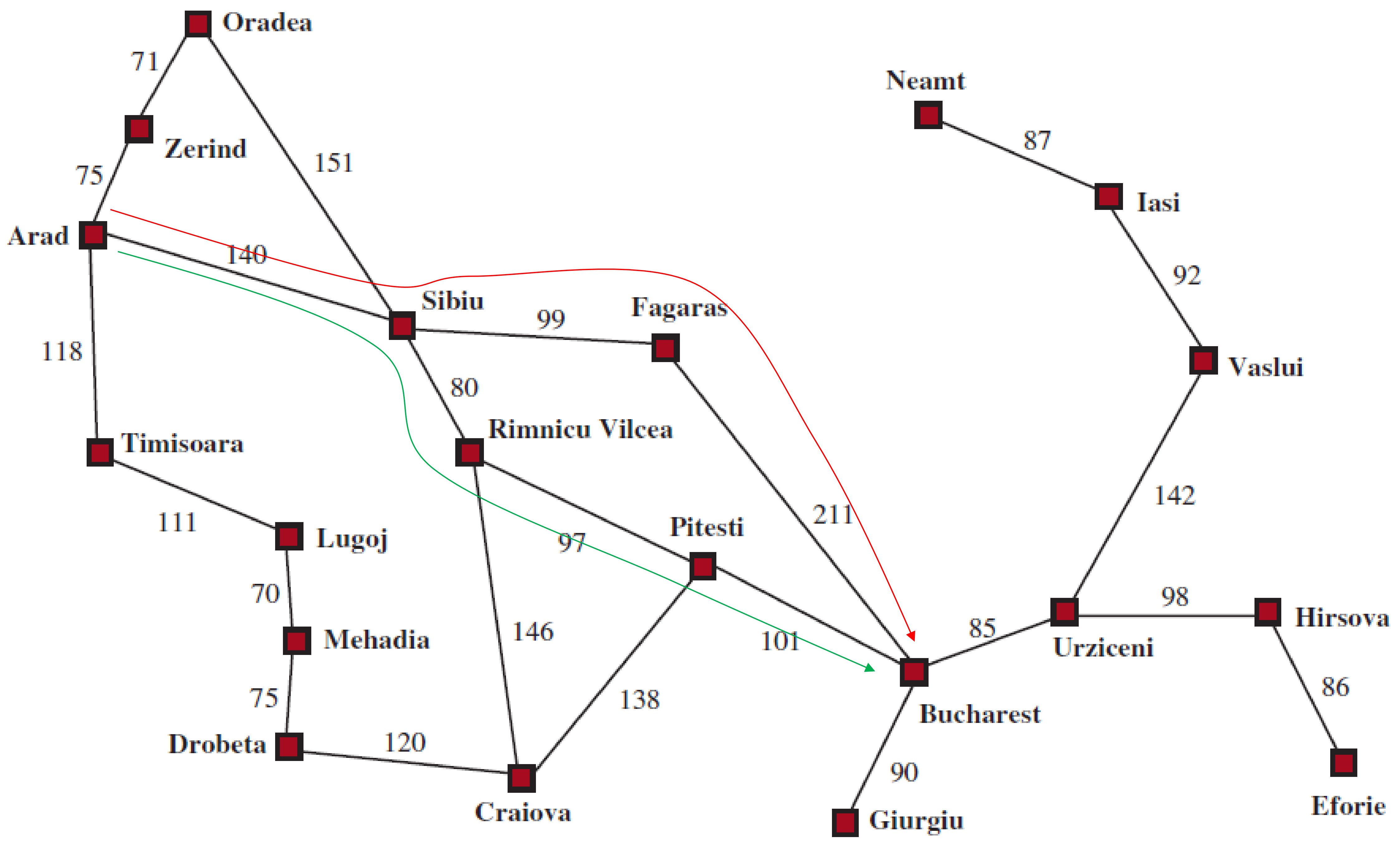




Greedy ≠ Optimal



Greedy solution: Arad – Sibiu – Fagaras – Bucharest (450)

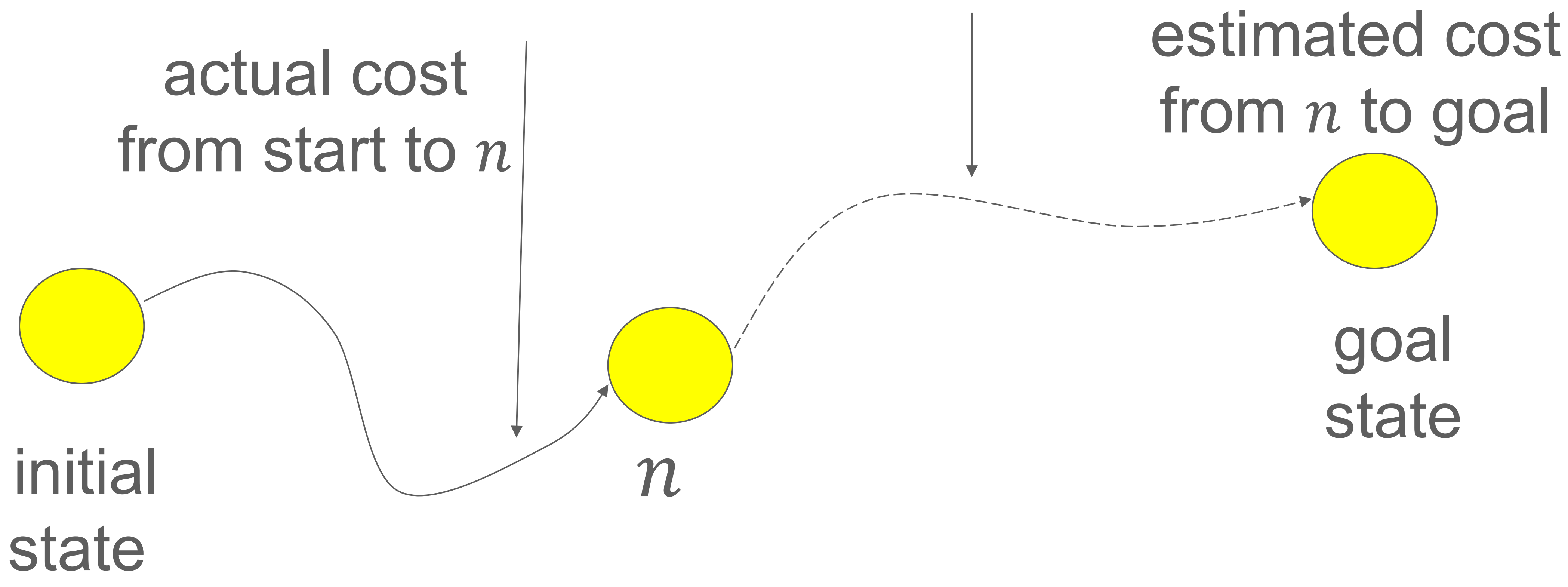
Optimal solution: Arad – Sibiu – Rimnicu Vilcea – Pitesti – Bucharest (418)

This greedy strategy does *not* account for the cost to get to the current state.

II. A* Search

P. E. Hart, N. J. Nilsson, and B. Raphael (1968)

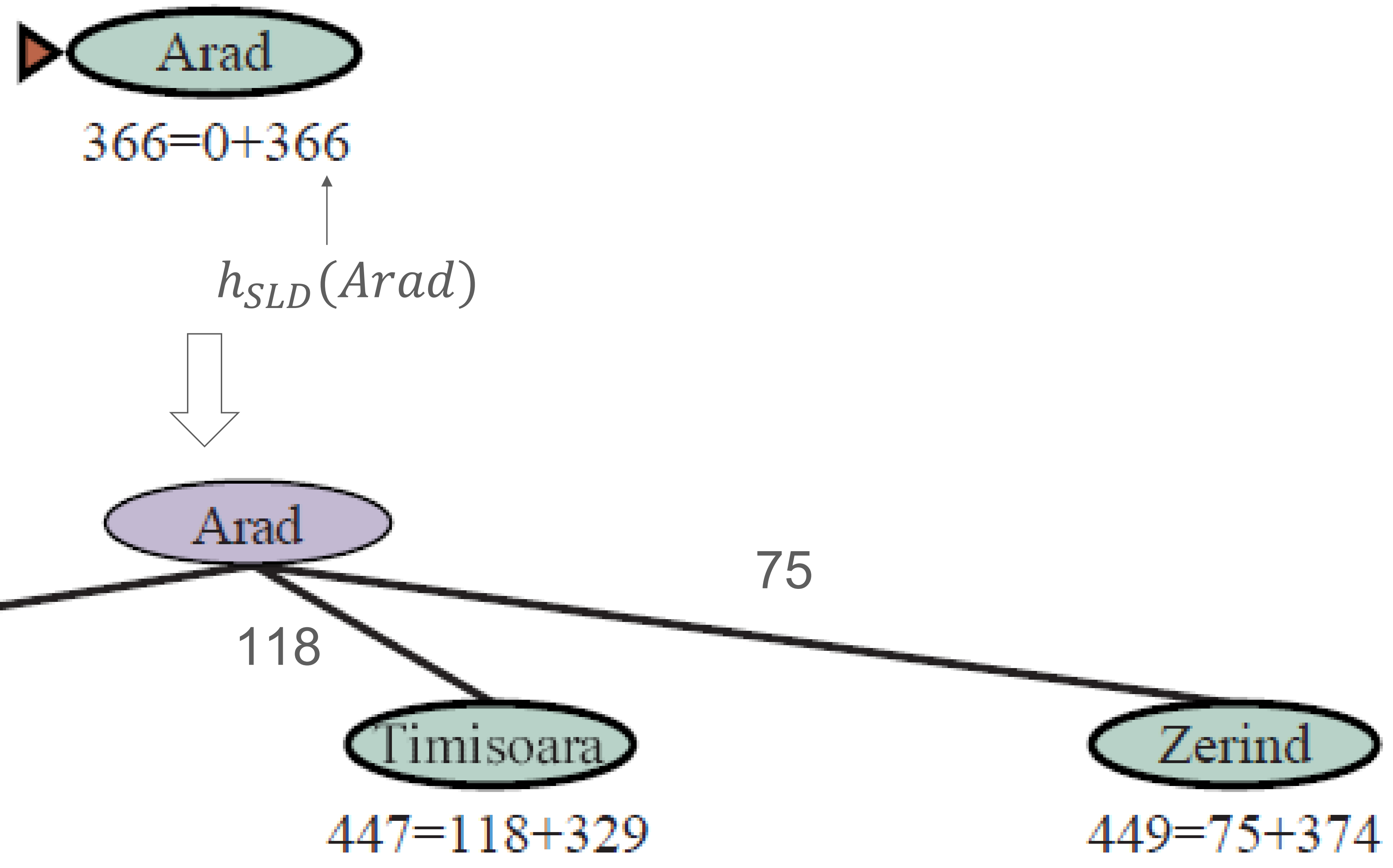
$$f(n) = g(n) + h(n)$$



$f(n)$: estimated cost of the best path that continues from n to a goal.

A* on Route Planning

$f(n) = g(n) + h_{SLD}(n)$



h_{SLD}: straight-line distance

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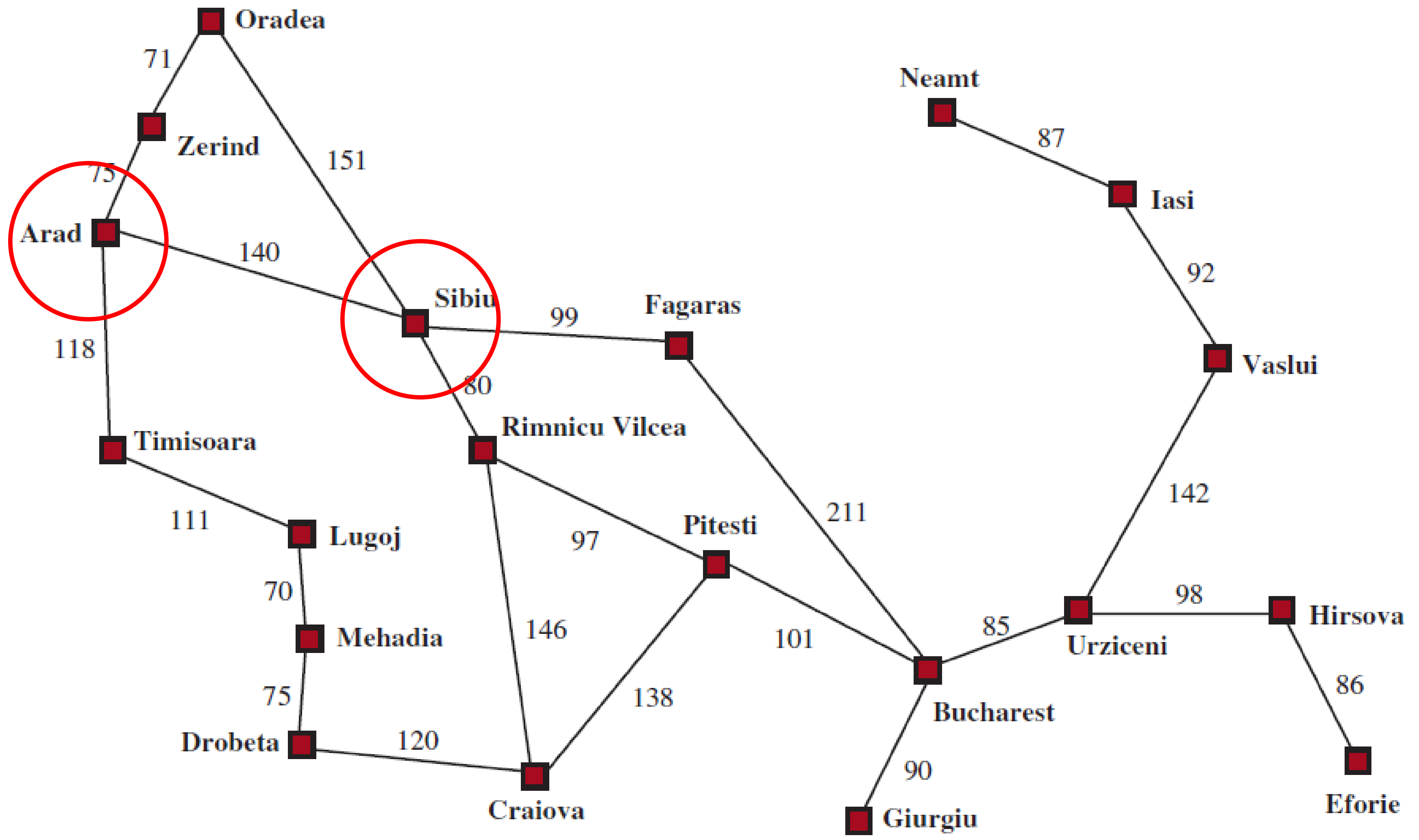
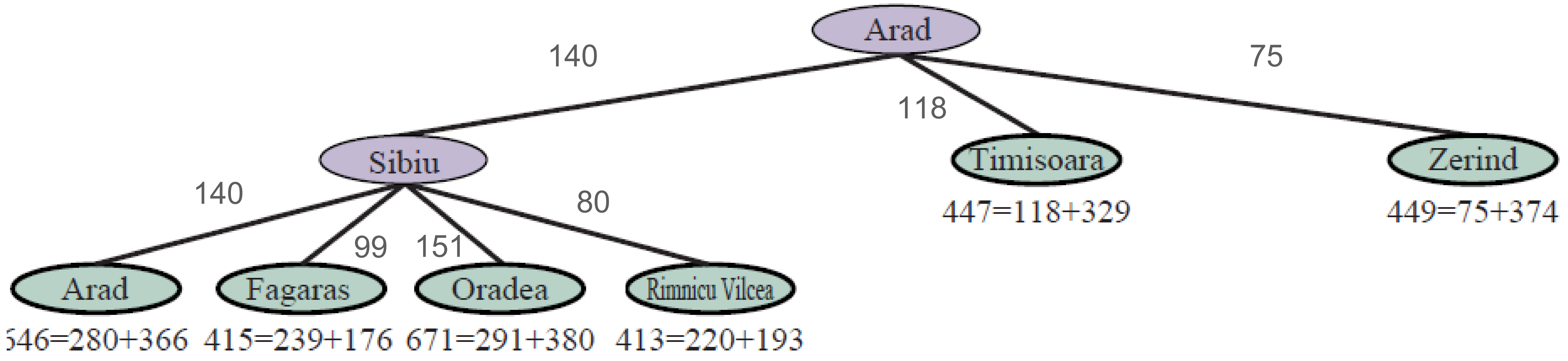


Working of A* Search



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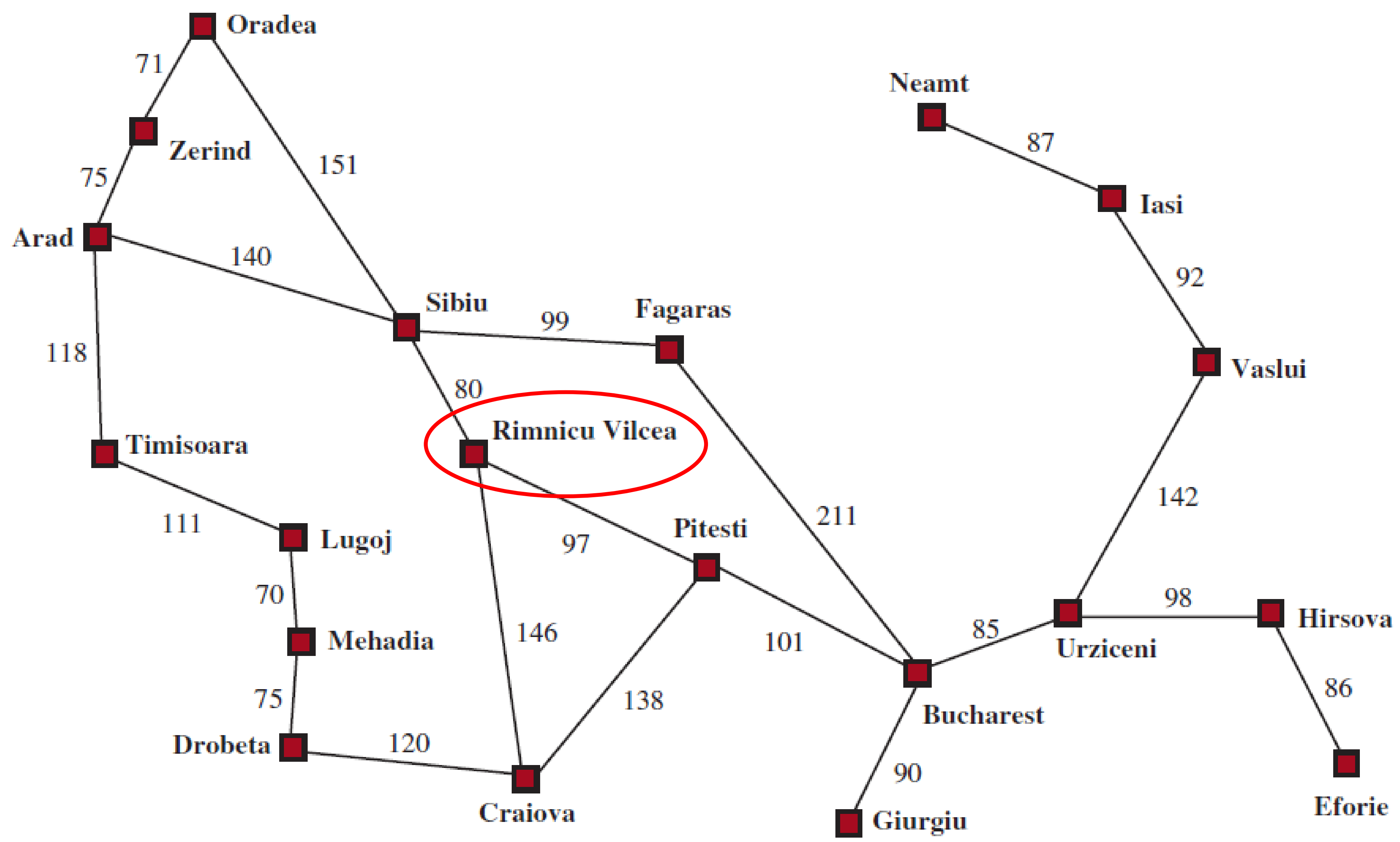

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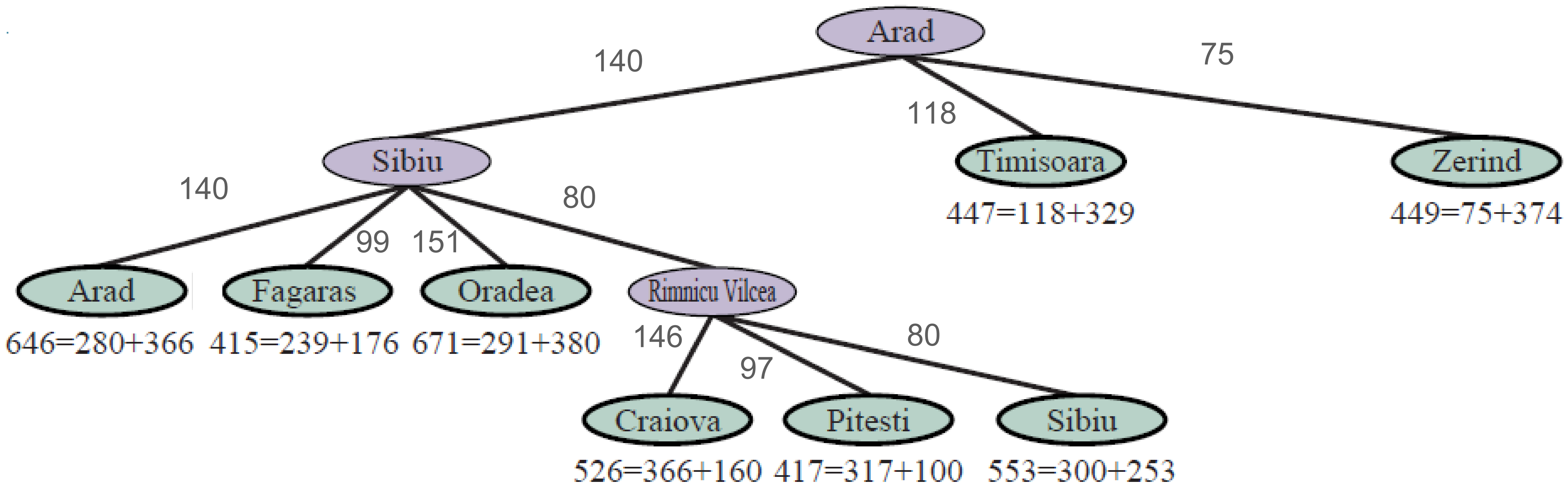

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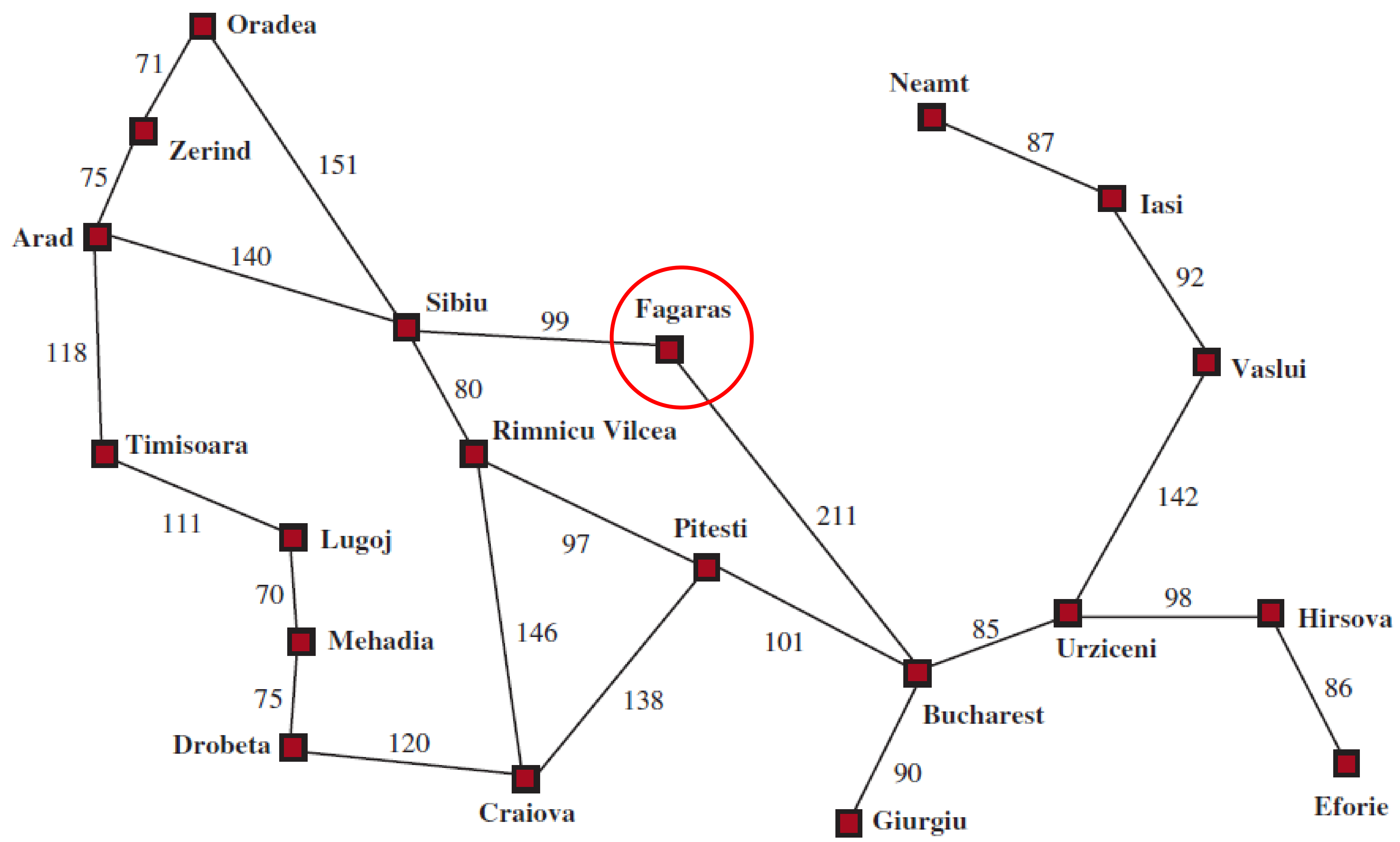

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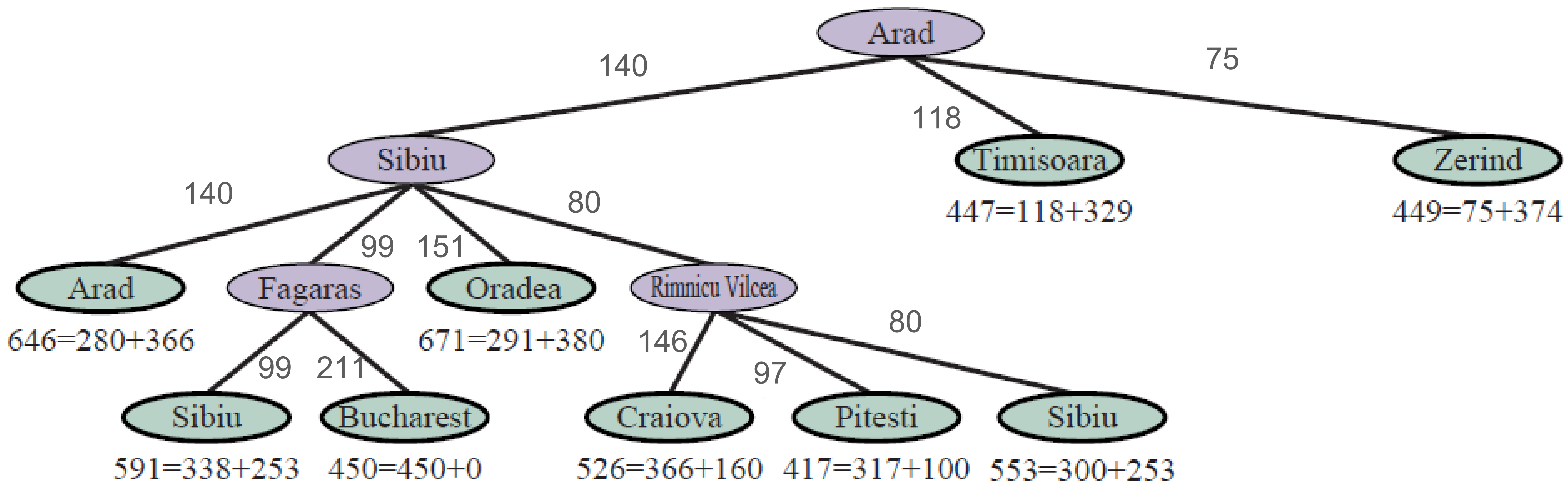

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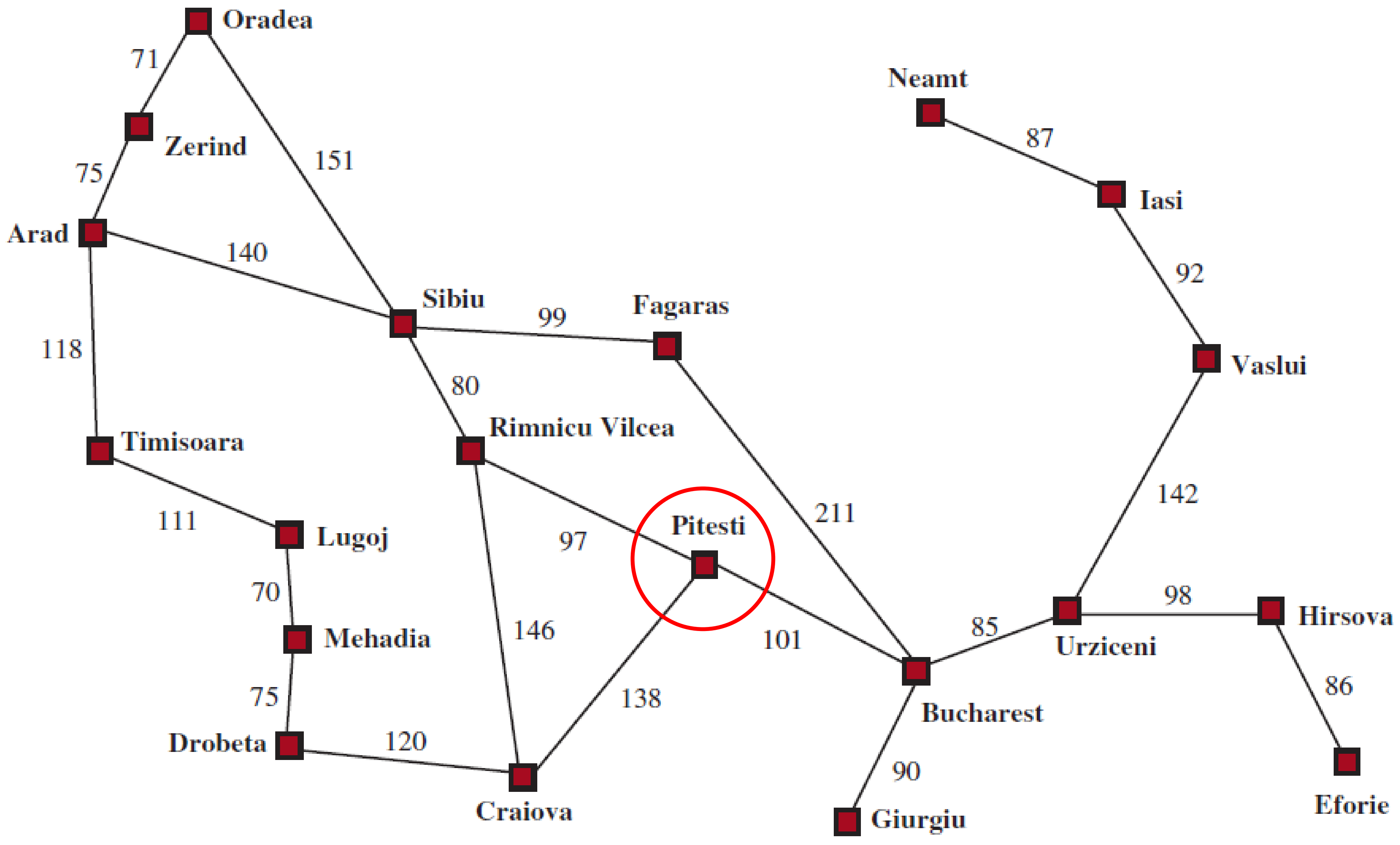
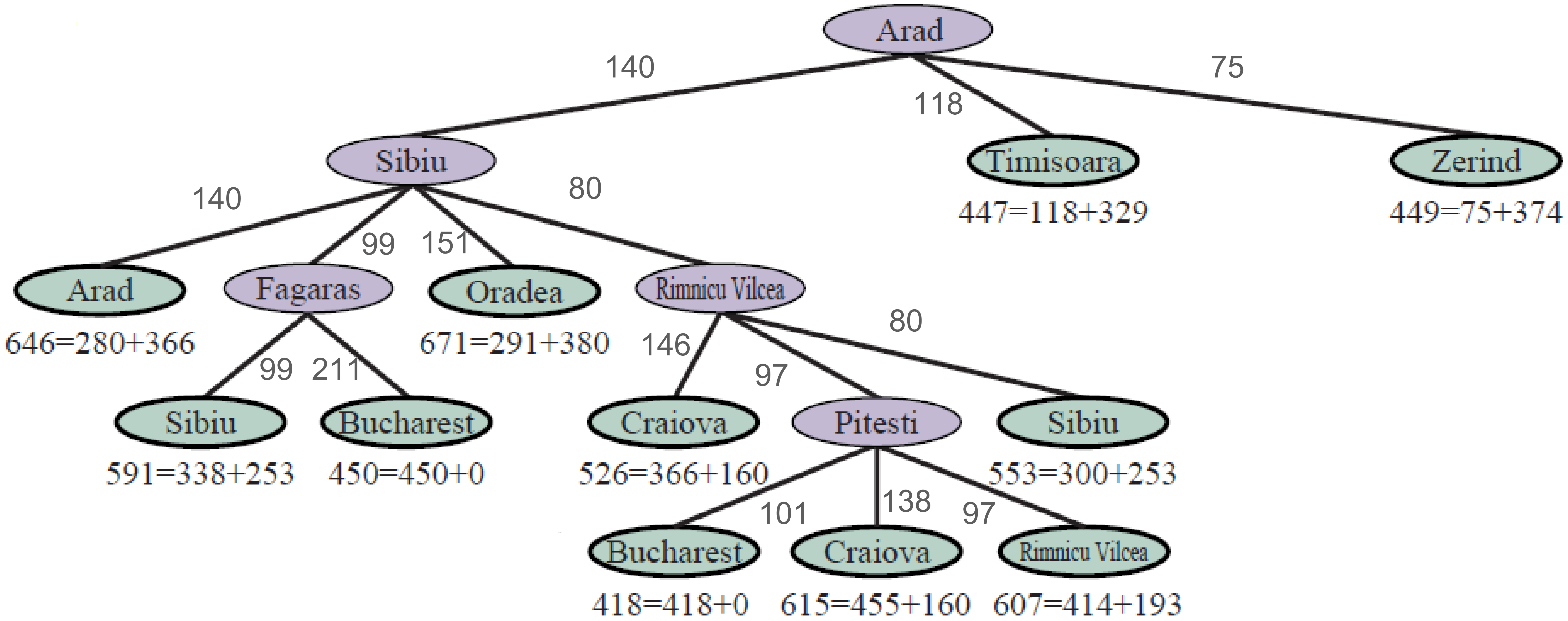


Working of A*



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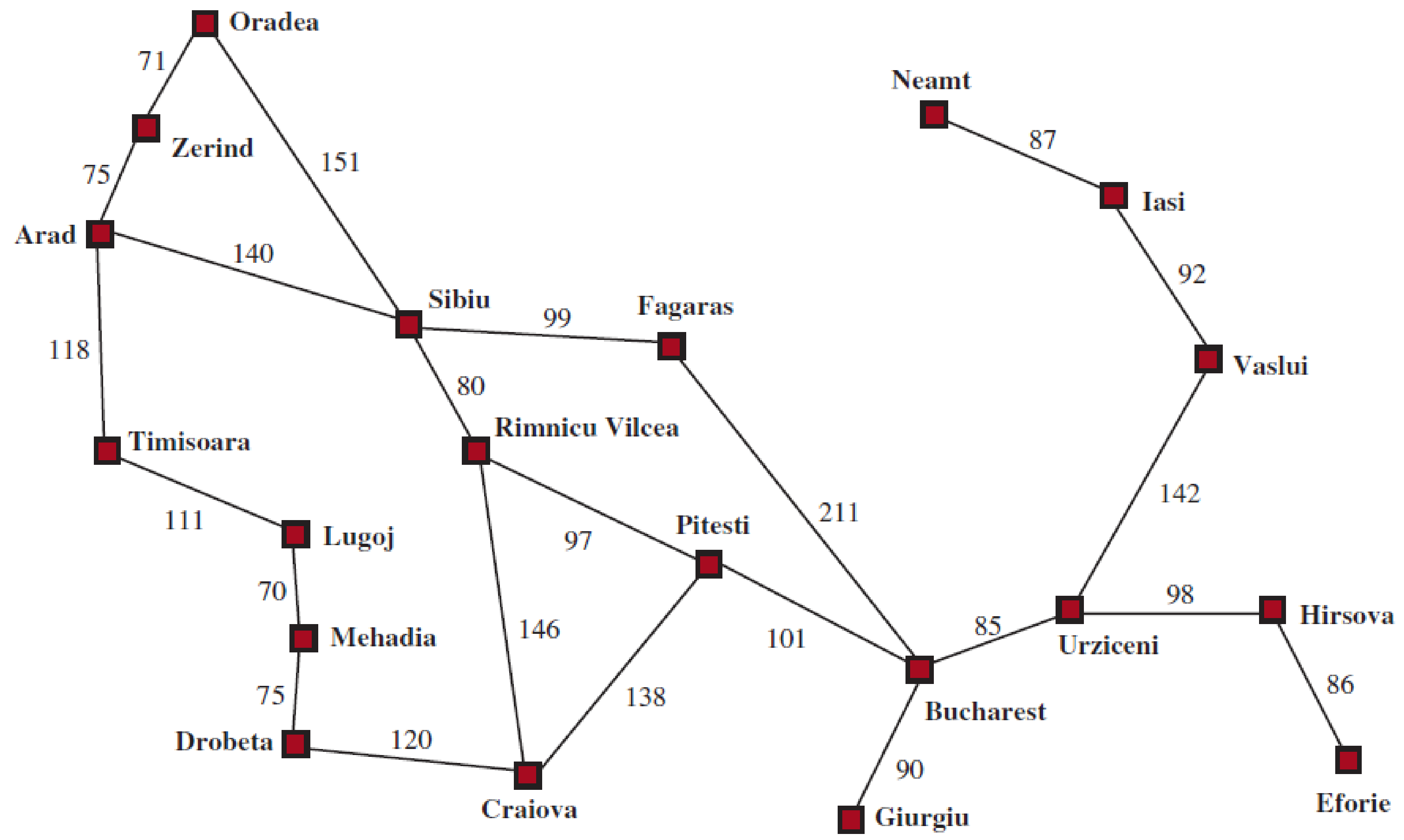

Properties of Heuristics

- A* search is *complete* (when the state space either has a solution or is finite).

- Whether it is optimal depends on the heuristic.

A heuristic is *admissible* if it *never overestimates* the cost to reach a goal.

h_{SLD}: straight-line distance

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h_{SLD} is admissible because the actual distance to Bucharest cannot be less than the straight-line distance.

Example (Best-first Search)

Best First Search

- Uniform-Cost Search: $f(n) = g(n)$
- Greedy Best First Search: $f(n) = h(n)$
- A* Search: $f(n) = g(n) + h(n)$

$g(n)$ = The actual cost from the initial state to the current state

$h(n)$ = Estimated cost of the cheapest path from the state at node n to a goal state

Greedy search vs. A* search

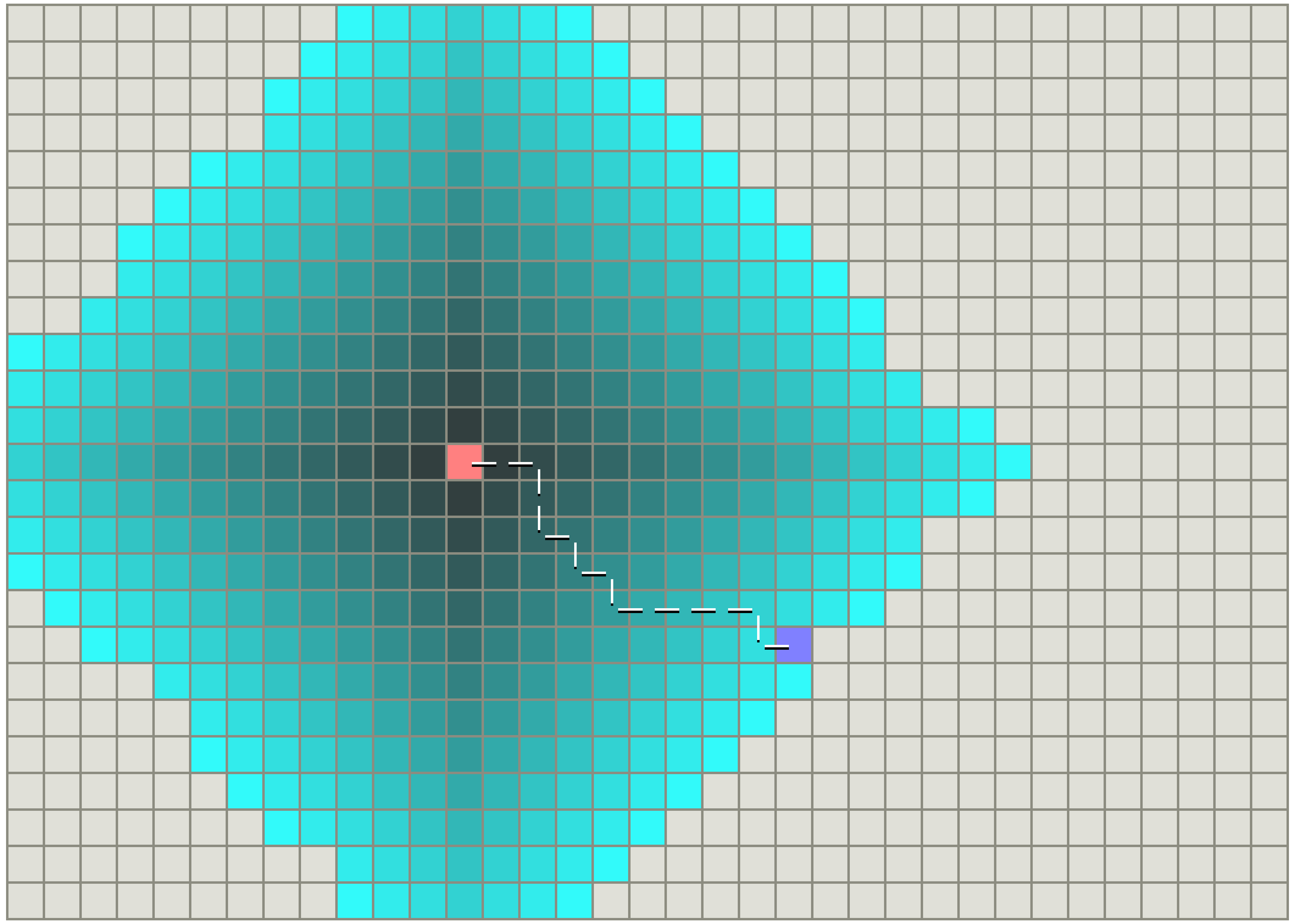
Greedy search

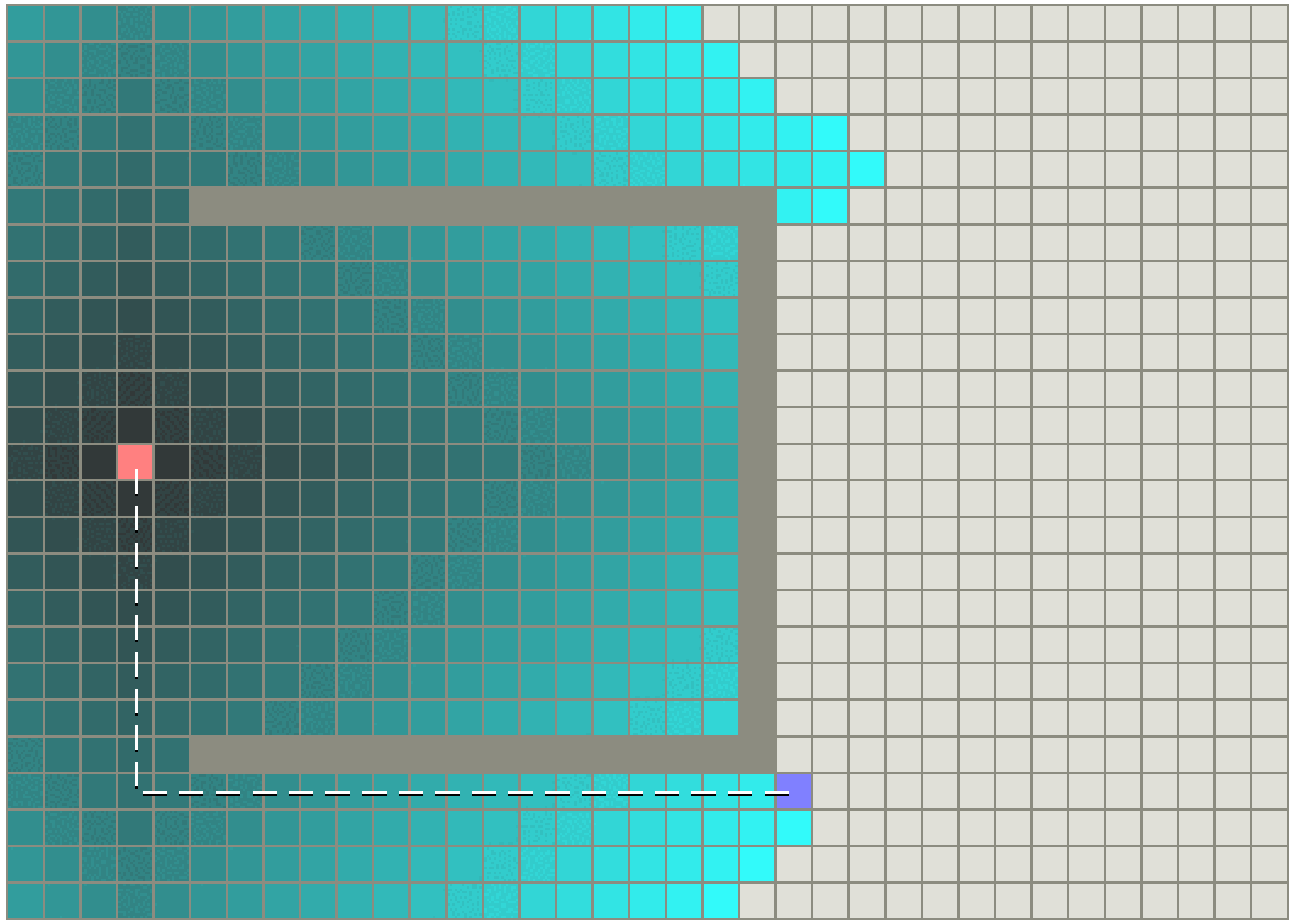
Criteria	Performance
Completeness	Complete with finite state space graph search, otherwise, not complete
Optimality	No
Time Complexity	Tree search $O(b^m)$ Graph search: size of state space
Space Complexity	Keep all nodes in memory Same as time complexity

A* search

Criteria	Performance
Completeness	Complete with finite state space graph search, otherwise, not complete
Optimality	Yes, if the heuristic function is admissible
Time Complexity	Tree search $O(b^m)$ Graph search: size of state space
Space Complexity	Keep all nodes in memory Same as time complexity

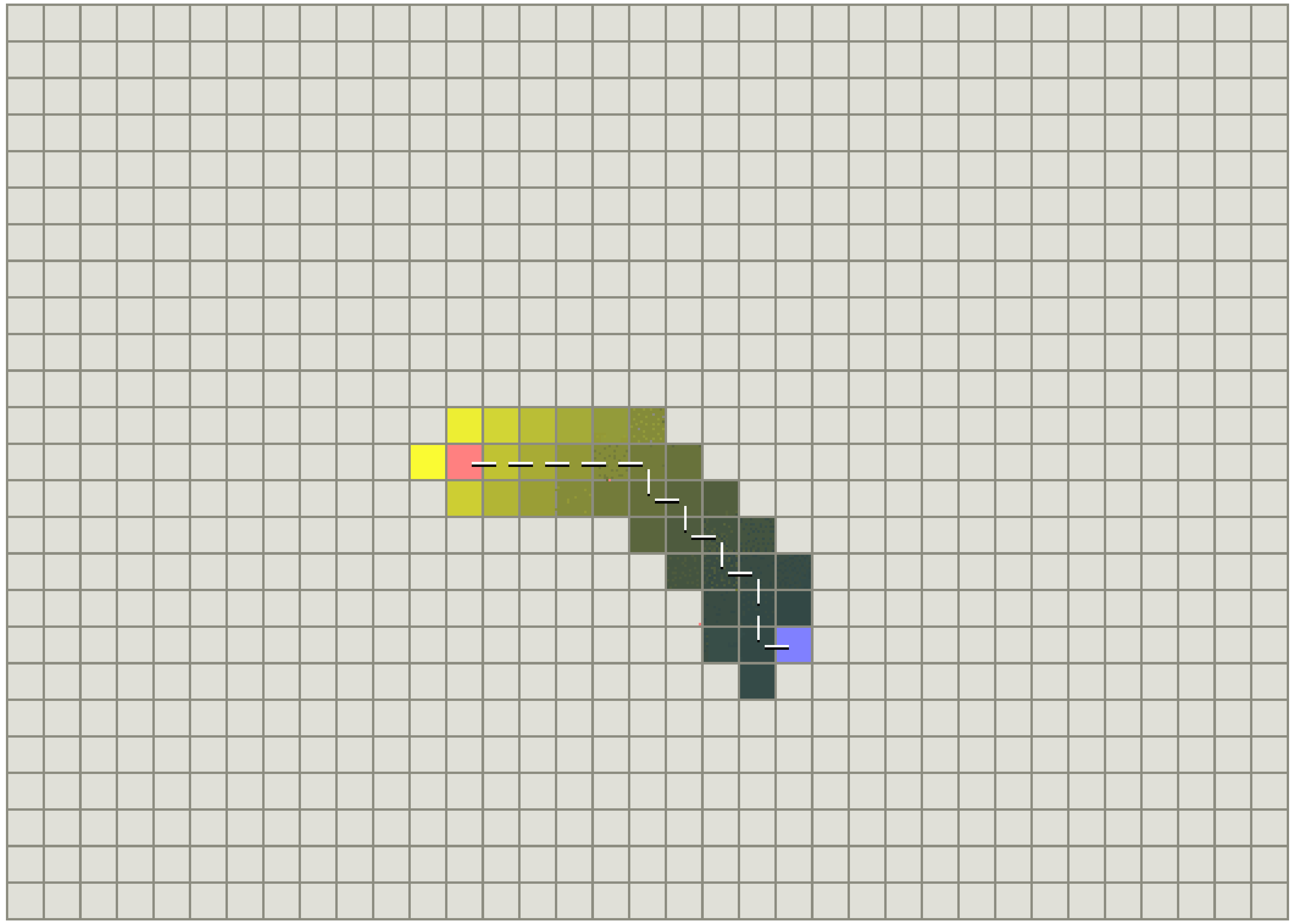
Example (Uniform Cost search)

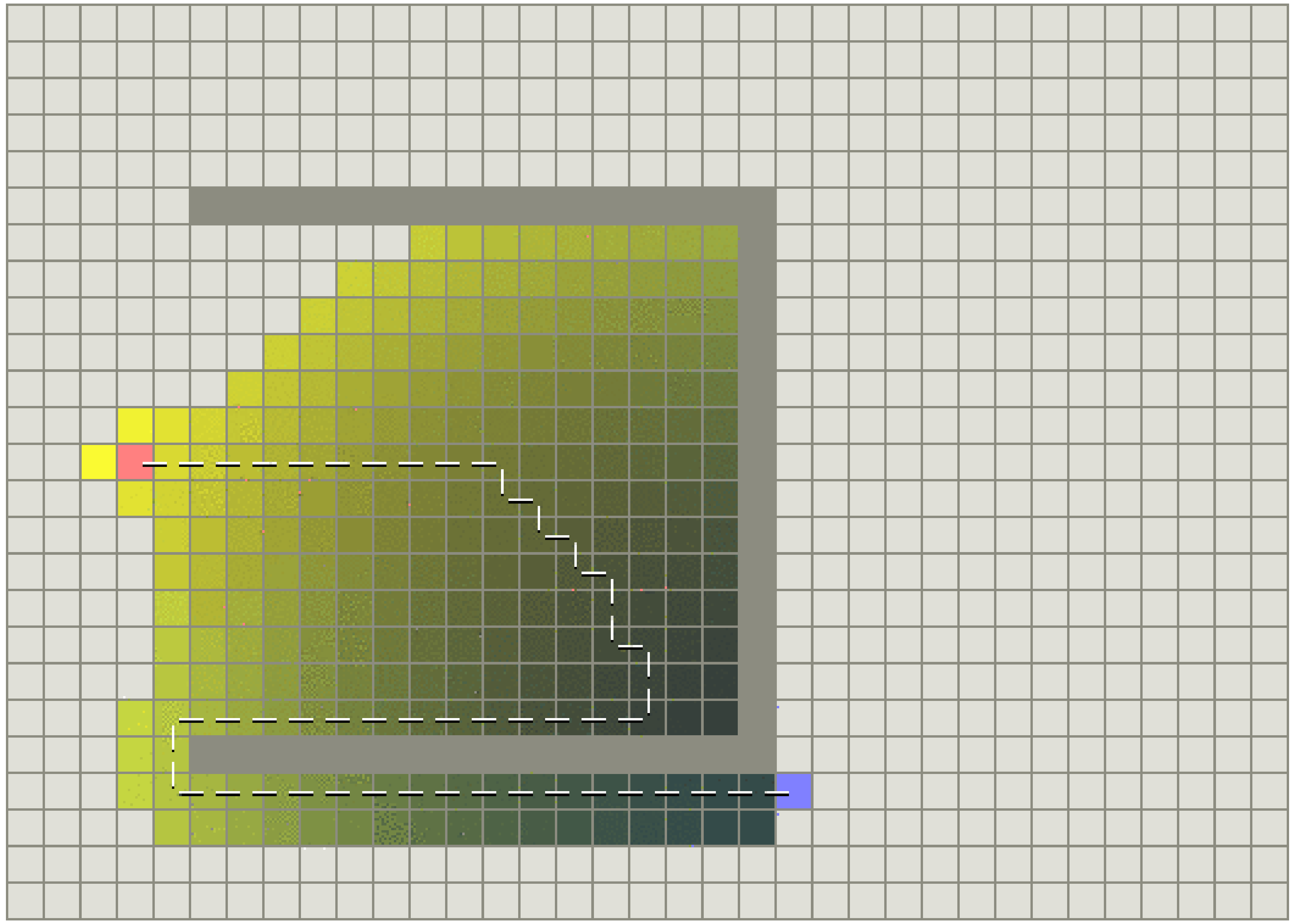




Ref:https://theory.stanford.edu/~amitp/GameProgramming/AStarComparison.html

Example (Greedy best first search)





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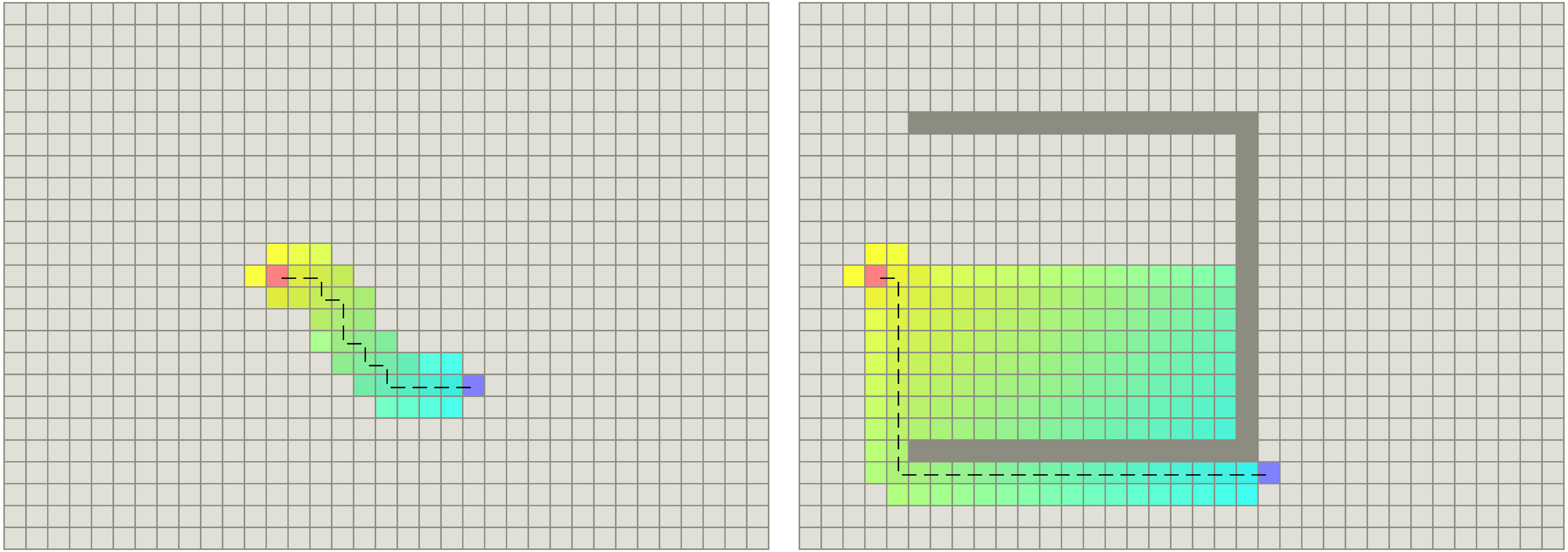
Example (A* search with admissible h)



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Example (Approximation Heuristics)

Examples of Heuristics:
- Manhattan Distance (on grid allows to move only in four directions only)
- Diagonal Distance (on grid allows to move in eight directions only)
- Euclidean Distance (on grid allows to move in any directions)

Which one is the best for this problem?
- All the heuristics are admissible since they will never overestimate the cost
- Manhattan distance is the best since it is closer to the actual cost than the other 2
- A* will expand less nodes, makes the search faster


That's all Folks!